

## Static Magnetic Fields

When a small test charge  $q$  is placed in an electric field  $\vec{E}$ , it experiences an electric force  $\vec{F}_e$ , which is a function of the position of  $q$ .

$$\vec{F}_e = q \vec{E} \quad (\text{N}) \quad \text{--- (i)}$$

In this eqn(i)  $\vec{E}$  is defined in terms of force for a "stationary charge".

When the test charge is in motion in a magnetic field (magnetic field is defined or characterized by, magnetic flux density  $\vec{B}$ ), "Experiments" show that charge  $q$  also experiences a "magnetic force  $\vec{F}_m$ ", given by

$$\vec{F}_m = q \vec{u} \times \vec{B} \quad (\text{N}).$$

Here, magnetic flux density defined in terms of force experienced by a "moving charge".

where,  $\vec{u}$  (m/s)  $\rightarrow$  velocity of moving charge

and,  $\vec{B} \rightarrow (\text{Wb/m}^2)$  or teslas (T).



Then, the total electromagnetic force on a charge  $q$  is, then, (2)

$$\vec{F} = \vec{F}_e + \vec{F}_m,$$

$$\Rightarrow \vec{F} = q(\vec{E} + \vec{u} \times \vec{B}) \quad (\text{N})$$

which is called Lorentz's force equation.

↗  
[Its validity has been established unquestionably by experiments.]



5-2: Write down the fundamental postulates of Magnetostatics in nonmagnetic media. (3)

Ans:- To study magnetostatics (steady magnetic fields) in free space or in non-magnetic material, we need only consider the magnetic flux density vector,  $\vec{B}$ .

The two fundamental postulates of magnetostatics that specify the divergence and the curl of  $\vec{B}$  in nonmagnetic media are

$$\nabla \cdot \vec{B} = 0$$

$$\text{and, } \nabla \times \vec{B} = \mu_0 \vec{J}$$

where,  $\mu_0 = 4\pi \times 10^{-7} \text{ (H/m)}$  and  $\vec{J}$  is current density  
(A/m<sup>2</sup>)

In integral form,

$$\oint_S \vec{B} \cdot d\vec{S} = 0$$

$$\text{and, } \oint_C \vec{B} \cdot d\vec{l} = \mu_0 I$$



pg-174 State Ampere's circuital law in nonmagnetic media. (4)

Ans: From the fundamental postulates we know,

$$\nabla \times \vec{B} = \mu_0 \vec{J}$$

Now, taking integral form, by integrating both sides over an open surface and applying Stokes' theorem, we have.

$$\int_S (\nabla \times \vec{B}) \cdot d\vec{S} = \mu_0 \int_S \vec{J} \cdot d\vec{S}$$

$$\Rightarrow \oint_C \vec{B} \cdot d\vec{r} = \mu_0 I \quad \text{--- (1)}$$

where, the path  $C$  for the line integral is the contour bounding the surface  $S$ , and  $I$  is the total current through  $S$ .

Eqn (1) is the form of Ampere's circuital law, which states that, "the circulation of the magnetic flux density in a nonmagnetic medium around any closed path is equal to  $\mu_0$  times the total current flowing through the surface bounded by the path."

→ This is very useful law in determining  $\vec{B}$  by a current  $I$ .



(5)

Example 5-1: An infinitely long, straight, solid, nonmagnetic conductor with a circular cross section of radius  $b$  carries a steady current  $I$ . Determine the magnetic flux density both inside and outside the conductor.

Ans: This is a problem with cylindrical symmetry and that Ampere's circuital law can be used to advantage.

a) Inside the conductor:

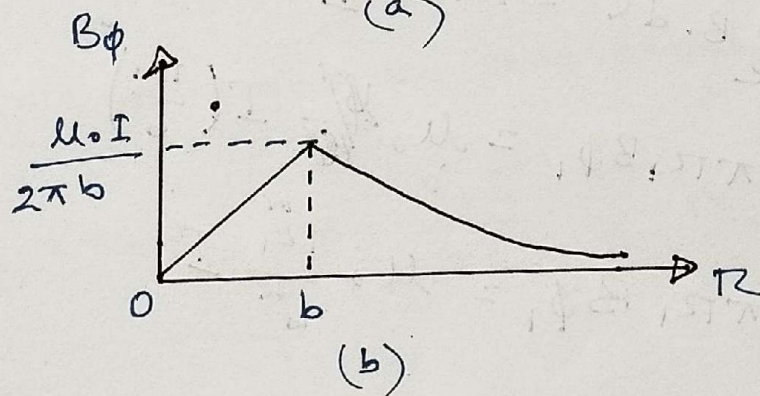
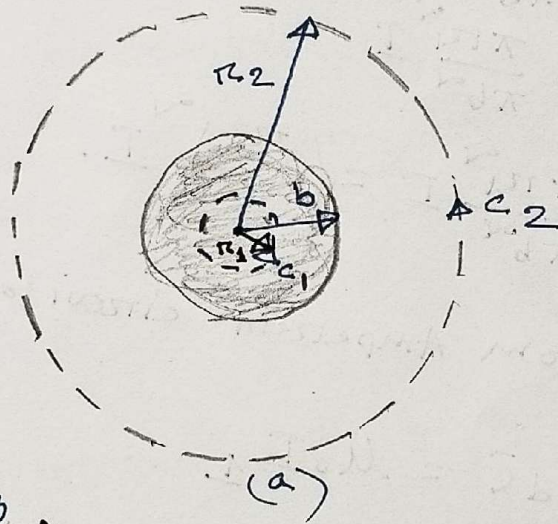
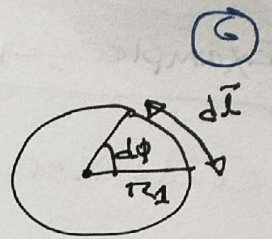


Fig. 5-2: Magnetic flux density of an infinitely long circular conductor carrying a current  $I$  out of paper.



$$\vec{B}_1 = a_\phi B_{\phi 1}; \quad d\vec{\ell} = a_\phi r_1 d\phi$$



As,

$$\oint_{C_1} \vec{B}_1 \cdot d\vec{\ell} = \int_0^{2\pi} B_{\phi 1} r_1 d\phi = 2\pi r_1 B_{\phi 1}. \quad [\because a_\phi \cdot a_\phi = 1]$$

The current through the area enclosed by  $C_1$  is,

$$\pi b^2 \rightarrow I$$

$$1 \rightarrow \frac{I}{\pi b^2}$$

$$\pi r_1^2 \rightarrow \frac{\pi r_1^2}{\pi b^2} I$$

$$\Rightarrow I_1 = \frac{\pi r_1^2}{\pi b^2} I = \left(\frac{r_1}{b}\right)^2 I.$$

therefore, from Ampere's circuital law

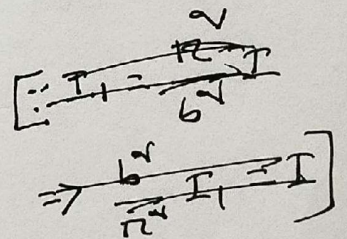
$$\oint_C \vec{B} \cdot d\vec{\ell} = \mu_0 I_1$$

$$\Rightarrow 2\pi r_1 B_{\phi 1} = \mu_0 \frac{\pi r_1^2}{b^2} I$$

$$\Rightarrow 2\pi r_1 B_{\phi 1} = \mu_0 \frac{r_1^2}{b^2} I$$

$$\Rightarrow B_{\phi 1} = \mu_0 \frac{r_1}{2\pi b^2} I$$

$$\Rightarrow \vec{B}_1 = a_\phi B_{\phi 1} = a_\phi \frac{\mu_0 r_1 I}{2\pi b^2}, \quad r_1 \leq b$$





(8)  $\therefore B_{\phi_1} = \text{at } b, \text{ i.e. } r_1 = b$

$$B_{\phi_1} = \mu_0 \frac{b}{2\pi b^2} I = \mu_0 \frac{1}{2\pi b} I$$

↓ maximum.

and,  $B_{\phi_1} \propto r_1$  [ $\because \mu_0, 2\pi, b, I$  are constant]

So that, 0 to b, line is linear and at b

$$B_{\phi_1} = \frac{\mu_0 I}{2\pi b}$$

b) Outside the conductor

$$\vec{B}_2 = \vec{a}_\phi B_{\phi_2}; \quad d\vec{\ell} = \vec{a}_\phi r_2 d\phi$$

$$\text{An, } \oint_{C_2} \vec{B}_2 \cdot d\vec{\ell} = 2\pi r_2 B_{\phi_2}$$

In this case total current is  $I$ , since  $C_2$  <sup>Path</sup> outside the conductor encloses the total current  $I$ .

$$\text{Hence, } \vec{B}_2 = \vec{a}_\phi B_{\phi_2} = \vec{a}_\phi \frac{\mu_0 I}{2\pi r_2}, \quad r_2 \geq b$$

$$\Rightarrow \boxed{B_{\phi_2} \propto \frac{1}{r_2}}$$

[ $B = \frac{k}{r}$ , is a hyperbola-type curve]



5-3 Define vector potential. Derive the vector Poisson's equation and write down its equivalent in Cartesian coordinates. (8)

Ans:- As in electrostatics, we obtain curl free  $\vec{E}$  from the gradient of electric potential  $V$ , as

$$\vec{E} = -\nabla V,$$

Similarly, divergence free magnetic field density  $\vec{B}$  can be expressed as the curl of another vector, say  $\vec{A}$ , such that

$$\vec{B} = \nabla \times \vec{A} \quad (\text{T})$$

The vector field  $\vec{A}$ , is called vector magnetic potential. Its SI unit is  $(\text{Wb/m})$ .

Now, we have from the fundamental postulates of magnetostatics in nonmagnetic media

$$\nabla \times \vec{B} = \mu_0 \vec{J}$$

$$\Rightarrow \nabla \times \nabla \times \vec{A} = \mu_0 \vec{J}$$

We know that,  $\nabla \times \nabla \times \vec{A} = \nabla(\nabla \cdot \vec{A}) - \nabla^2 \vec{A}$

$$\Rightarrow \nabla^2 \vec{A} = \nabla(\nabla \cdot \vec{A}) - \nabla \times \nabla \times \vec{A}$$



(3)

Here,  $\nabla^2 \bar{A}$ , is the Laplacian of  $\bar{A}$ .

For cartesian coordinates it can also be defined

$$\nabla^2 \bar{A} = \bar{a}_x \nabla^2 A_x + \bar{a}_y \nabla^2 A_y + \bar{a}_z \nabla^2 A_z.$$

Now, we can write

$$\nabla(\nabla \cdot \bar{A}) - \nabla^2 \bar{A} = \mu_0 \bar{J}$$

For simplicity, from the Coulomb definition  
condition for divergence of  $\bar{A}$ ,

$$\nabla \cdot \bar{A} = 0$$

$$\left[ \begin{array}{l} \text{as, } \nabla \cdot \bar{B} = 0 \\ \therefore, \nabla \cdot \bar{A} = 0 \end{array} \right]$$

$$\text{then, } \boxed{\nabla^2 \bar{A} = -\mu_0 \bar{J}}$$

This is a operator form of Vector Poisson's equation.

In cartesian coordinates,

$$\nabla^2 A_x = -\mu_0 J_x,$$

$$\nabla^2 A_y = -\mu_0 J_y,$$

$$\nabla^2 A_z = -\mu_0 J_z.$$



5-4% Deduce the expression of Biot-Savart (10)  
law for finding magnetic flux from current  
in a closed circuit.

Ans:- In many applications we are interested  
in determining the magnetic field due to a  
a current-carrying circuit.

For a thin wire with cross-sectional  
area  $S$ , and for small incremental length  $d\ell'$ ,  
we can write, the current flow is entirely  
along the wire as,

$$\vec{J} dV' = JS d\ell' = I d\ell' \quad \text{--- (i)}$$

Now, As we know vector magnetic  
potential  $\vec{A}$  from current in a  
closed circuit,

$$\vec{A} = \frac{\mu_0}{4\pi} \int_{V'} \frac{\vec{J}}{R} dV' \quad (\text{Wb/m}) \quad \text{--- (ii)}$$

[ $\because$  it is similar as,

$$V = \frac{1}{4\pi\epsilon_0} \int_{V'} \frac{\rho_v}{R} dV']$$

$$[\because dV' = S d\ell']$$

$$[\because JS = I]$$

$\downarrow$  current density  
 $\downarrow$  area

$$[\because J \times dV' \rightarrow (A/m^2) \times (m^3)]$$

$$[= A \cdot m]$$

$\hookrightarrow$  it represents  
current element  
not current

$$[\because I = \int \vec{J} \cdot d\vec{S}]$$

$$J \times \text{area} = (A/m^2) \times m^2 = A$$

it is current

$$[\because I d\ell = A \cdot m]$$



Now, from (i) we can obtain  $\vec{A}$  as,

$$\vec{A} = \frac{\mu_0}{4\pi} \oint_{C'} \frac{I d\vec{\ell}'}{R} \quad (\text{wb/m}) \quad \dots \dots (iii)$$

The magnetic field density is then

$$\vec{B} = \nabla \times \vec{A} = \nabla \times \left[ \frac{\mu_0 I}{4\pi} \oint_{C'} \frac{d\vec{\ell}'}{R} \right]$$

$$= \frac{\mu_0 I}{4\pi} \oint_{C'} \nabla \times \left( \frac{d\vec{\ell}'}{R} \right) \quad \dots \dots (iv)$$

from mathematics we know that

$$\nabla \times (f \vec{G}) = f \nabla \times \vec{G} + (\nabla f) \times \vec{G}$$

Here,  $f = \frac{1}{R}$  and  $\vec{G} = d\vec{\ell}'$ , then,

$$\vec{B} = \frac{\mu_0 I}{4\pi} \oint_{C'} \left[ \frac{1}{R} \nabla \times d\vec{\ell}' + \left( \nabla \frac{1}{R} \right) \times d\vec{\ell}' \right] \quad \dots \dots (v)$$

Now, here, we consider  $d\vec{\ell}'$  as small element of source wire, so we consider the  $(x', y', z')$  as the coordinates of source point, these are called primed coordinates.



+

And, consider the coordinates of field point where we want to find the field  $\vec{B}$  as  $(x, y, z)$ , these are called as "unprimed coordinates"

+ Now, from these primed and unprimed coordinates

we can write,

$$\nabla \times d\vec{r}' = 0$$

↑

because,

$$\nabla = \vec{a}_x \frac{\partial}{\partial x} + \vec{a}_y \frac{\partial}{\partial y} + \vec{a}_z \frac{\partial}{\partial z}$$

and  $d\vec{r}' = \vec{a}_x dx' + \vec{a}_y dy' + \vec{a}_z dz'$

so,

$$\frac{\partial}{\partial x} (dx') = 0 \quad [\because \text{Here, } x \text{ and } x' \text{ are different variables}]$$

then,

$$\nabla \times d\vec{r}' = 0$$

So, in eq<sup>n</sup> (v), the first term is vanished.

then, in (v),  $R \rightarrow r$  is the distance from

field point to the source point,

Ans,

$$R = |\vec{r} - \vec{r}'|$$

where,

$$\vec{r} = (x, y, z)$$

$$\vec{r}' = (x', y', z')$$



then, we know,

$$R = \sqrt{(x-x')^2 + (y-y')^2 + (z-z')^2}$$

$$\Rightarrow \frac{1}{R} = \left[ (x-x')^2 + (y-y')^2 + (z-z')^2 \right]^{-1/2}$$

then,

$$\nabla \left( \frac{1}{R} \right) = \bar{a}_x \frac{\partial}{\partial x} \left( \frac{1}{R} \right) + \bar{a}_y \frac{\partial}{\partial y} \left( \frac{1}{R} \right) + \bar{a}_z \frac{\partial}{\partial z} \left( \frac{1}{R} \right)$$

Now,

$$\frac{\partial}{\partial x} \left( \frac{1}{R} \right) = \frac{\partial}{\partial x} (R^{-1}) = -1 \cdot R^{-2} \cdot \boxed{\frac{\partial R}{\partial x}}$$

then,

$$\frac{\partial R}{\partial x} = \frac{\partial}{\partial x} \left[ (x-x')^2 + (y-y')^2 + (z-z')^2 \right]^{1/2}$$

$$= \frac{\partial}{\partial x} (u^{1/2})$$

$\therefore$  suppose,  
 $u = (x-x')^2 + (y-y')^2 + (z-z')^2$

$$\boxed{\frac{\partial R}{\partial x}} = \frac{1}{2} u^{-1/2} \boxed{\frac{\partial u}{\partial x}}$$

then,

$$\frac{\partial u}{\partial x} = \frac{\partial}{\partial x} \left[ (x-x')^2 + (y-y')^2 + (z-z')^2 \right]$$

$$= \frac{\partial}{\partial x} (x-x')^2 + 0 + 0$$

$$= 2(x-x') \left[ \frac{\partial}{\partial x} (x) - \frac{\partial}{\partial x} (x') \right]$$

$$= 2(x-x') [1 - 0]$$

$$\boxed{\frac{\partial u}{\partial x}} = 2(x-x')$$



then,  $\frac{\partial R}{\partial x} = \frac{1}{2} u^{-1/2} \cdot 2(x-x')$

$$= (x-x') u^{-1/2}$$

Here,  $u^{1/2} = R$

$$\left[ \because R = \sqrt{u} \right. \\ \left. = u^{1/2} \right]$$

$$\Rightarrow u^{-1/2} = \frac{1}{R}$$

Finally,  $\boxed{\frac{\partial R}{\partial x} = \frac{x-x'}{R}}$

Now,  $\frac{\partial}{\partial x} \left( \frac{1}{R} \right) = -R^{-2} \cdot \frac{x-x'}{R} = -\frac{x-x'}{R^3}$

Similarly,  $\frac{\partial}{\partial y} \left( \frac{1}{R} \right) = -\frac{y-y'}{R^3}$

$$\frac{\partial}{\partial z} \left( \frac{1}{R} \right) = -\frac{z-z'}{R^3}$$

So,  $\nabla \left( \frac{1}{R} \right) = -\frac{(x-x')\vec{a}_x + (y-y')\vec{a}_y + (z-z')\vec{a}_z}{R^3}$

As,  $\vec{R} = (x-x')\vec{a}_x + (y-y')\vec{a}_y + (z-z')\vec{a}_z$

then,  $\nabla \left( \frac{1}{R} \right) = -\frac{\vec{R}}{R^3} = -\frac{R\vec{a}_R}{R^3} = -\frac{\vec{a}_R}{R^2}$



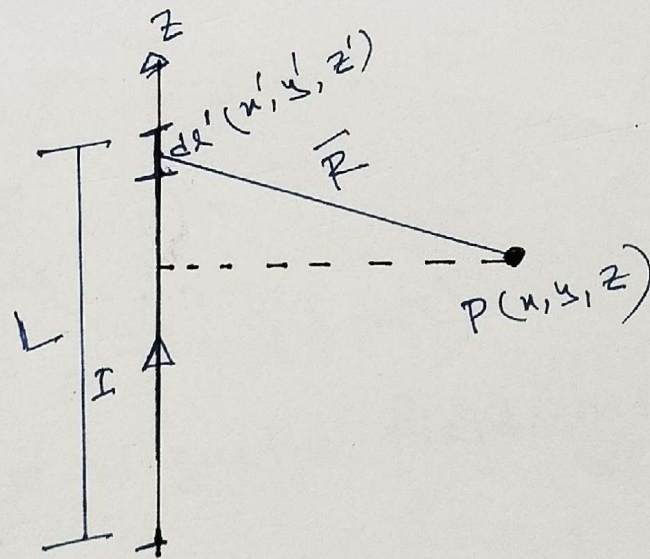
then, from (v)  $\Rightarrow$

15

$$\vec{B} = \frac{\mu_0 I}{4\pi} \oint_{C'} \left( -\vec{a}_R \frac{1}{R^2} \right) \times d\vec{\ell}' \quad (T)$$

$$\text{then, } \vec{B} = \frac{\mu_0 I}{4\pi} \oint_{C'} \frac{d\vec{\ell}' \times \vec{a}_R}{R^2} \quad (T)$$

this is the Biot-Savart law, for determining the magnetic flux density from current in a closed circuit.





Now, if,  $\vec{m}_k$  be the magnetic dipole moment of an atom,

then, magnetization vector,  $\vec{M}$  is,

$$\vec{M} = \lim_{\Delta v \rightarrow 0} \frac{\sum_{k=1}^{n \Delta v} \vec{m}_k}{\Delta v} \quad (\text{A/m})$$

then, magnetization volume current density

$$\vec{J}_{mv} = \nabla \times \vec{M} \quad (\text{A/m}^2)$$

$$\text{then, } \frac{1}{\mu_0} \nabla \times \vec{B} = \vec{J} + \vec{J}_{mv}$$

$$[\because \nabla \times \vec{B} = \mu_0 \vec{J}]$$

$$\Rightarrow \frac{1}{\mu_0} \nabla \times \vec{B} = \vec{J} + \nabla \times \vec{M}$$

$$\Rightarrow \nabla \times \left( \frac{\vec{B}}{\mu_0} - \vec{M} \right) = \vec{J}$$

$$\Rightarrow \nabla \times \vec{H} = \vec{J} \quad (\text{A/m}^2)$$

$\vec{H}$  magnetic field intensity.

Thus,



ଏହା, ଆହୁରି କେବଳ-କାନ୍ଦି ଏ, ଯଦି externally  
 ଏହା- magnetic field apply କରି. ତେବେ conduc-  
 -ତା ଏ current flow ହେବା ଯାଏ ଏ  
 magnetic field  $B$  ଯାଏ ହେଉଛି, ତାହା-  
 ଯାଏ- ଯାଏ change ହେଉ, ଏହା ଏ external  
 magnetic field apply ଏହା ଯାଏ ଏହା ହେଉଛି  
 ଏହା- magnetizing vector ଯାଏ ହେଉଛି

$$[\vec{E} = -\vec{\nabla} \phi]$$

$$\vec{M} \times \vec{\nabla} + \vec{E} = -\vec{\nabla} \times \frac{\vec{B}}{\mu_0}$$

$$\vec{M} \times \vec{\nabla} + \vec{E} = -\vec{\nabla} \times \frac{\vec{B}}{\mu_0}$$

$$\vec{E} = \left( \vec{M} - \frac{\vec{B}}{\mu_0} \right) \times \vec{\nabla}$$

$$\vec{E} = \vec{H} \times \vec{\nabla}$$

(v/m) magnetic field intensity



Now, if we consider surface integral  
in both sides

$$\int_S (\nabla \times \vec{H}) \cdot d\vec{S} = \int_S \vec{J} \cdot d\vec{S}$$

then, according to the Stoke's theorem

$$\oint_C \vec{H} \cdot d\vec{l} = I \quad (A) \quad \dots$$

is the contour (closed path) and this eq<sup>n</sup> is the generalized form of Ampere's circuital law for steady current - applicable in non-magnetic as well as magnetic media.

\* Also, for linear and isotropic case

$$\vec{M} = \chi_m \vec{H}$$

where,  $\chi_m$  is magnetic susceptibility.

$$\text{then, } \vec{B} \text{ is from, } \vec{H} = \frac{\vec{B}}{\mu_0} - \vec{M} = \frac{\vec{B}}{\mu_0} - \chi_m \vec{H}$$

$$\Rightarrow \vec{B} = \mu_0 (1 + \chi_m) \vec{H}$$

$$\Rightarrow \vec{B} = \mu_0 \mu_r \vec{H} = \mu \vec{H} \quad (\text{wb/m}^2)$$



$$\Rightarrow \vec{H} = \frac{1}{\mu} \vec{B}$$

where,  $\mu_r = 1 + \chi_m = \frac{\mu}{\mu_0}$

$\mu$  → absolute permeability,  
 $\mu_0$  → permeability in free space.

↓  
relative permeability.

(A)

$$\vec{I} = \vec{J} \cdot \vec{H}$$



## 5-10 Inductances And Inductors

(19)

In static condition we describe capacitor effect, when current is steady then we can describe the effect of Inductance.

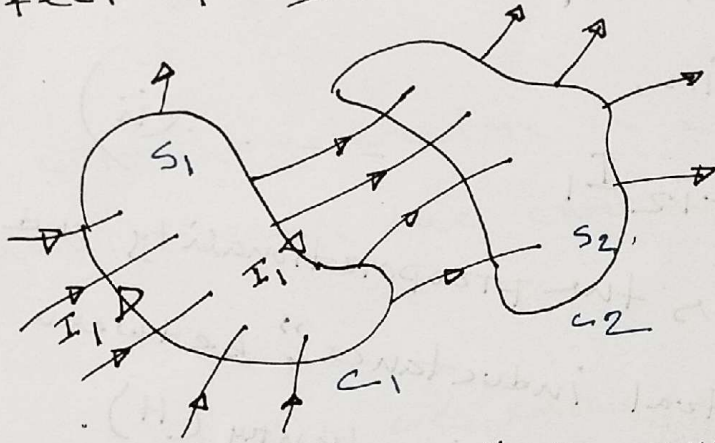


Fig. 5-14: Two magnetically coupled loops.

Considering,

Two neighboring closed loops  $C_1$  &  $C_2$  with bounding surface areas  $S_1$  &  $S_2$  respectively. If, a current  $I_1$  flows in  $C_1$ , a magnetic field  $\vec{B}_1$  will be created. Some of magnetic flux due to  $\vec{B}_1$ , will link with  $C_2$ .

Let, us designate this mutual flux as

$$\phi_{12} = \int_{S_2} \vec{B}_1 \cdot d\vec{S}_2 \quad (\text{Wb}) \quad \dots (i)$$



As we know from Biot-Savart law,

$$\vec{B}_1 = \frac{\mu_0 I_1}{4\pi} \oint_C \frac{d\vec{\ell} \times \vec{a}_R}{R^2} \quad (T)$$

we see that,  $B_1$  is directly proportional to  $I_1$ , that is,

$$B_1 = L_{12} I_1 \quad \dots \dots (ii)$$

where,  $L_{12}$  is the proportionality and it is called the "mutual inductance" between loops  $C_1$  &  $C_2$  with SI unit Henry (H).

$$\begin{aligned} \text{then, from (i)} \Rightarrow \phi_{12} &= \int_{S_2} L_{12} I_1 \cdot d\vec{S}_2 \\ &= L_{12} I_1 \int_{S_2} dS_2 \end{aligned}$$

$$\Rightarrow \boxed{\phi_{12} = L_{12} I_1}$$

OR simply,  $\boxed{\phi = L I}$   
 $\downarrow$   
 inductance.



5-11: Determine the magnetic energy stored (21) in two coupled current carrying loops.

Ans:- Consider a single closed loop with a self inductance  $L_1$ , in which the current is initially zero.

A current generator is connected to the loop, which increases the current  $i_1$  from zero to  $I_1$ .

From physics we know that an EMF will be induced in the loop that opposes the current change.

An amount of work must be done to overcome this induced emf.

Let,  $v_1 = L_1 \frac{di_1}{dt}$  be the voltage across the inductance.

the work required is

$$W_1 = \int v_1 i_1 dt = L_1 \int_0^{I_1} i_1 di_1 = \frac{1}{2} L_1 I_1^2 \quad \dots (1)$$

which is stored as magnetic energy.



(22)

Now consider two closed loops  $C_1$  and  $C_2$  carrying currents  $i_1$  and  $i_2$ , respectively.

To find the amount of work required, we first keep  $i_2 = 0$  and increase  $i_1$  from zero to  $I_1$ . This requires a work  $W_1$  in loop  $C_1$ , as given in eqn. (i). No work is done in loop  $C_2$ , since  $i_2 = 0$ .

Next we keep  $i_1$  at  $I_1$  and increase  $i_2$  from zero to  $I_2$ .

Because of mutual coupling, some of magnetic flux due to  $i_2$  will link with loop  $C_1$ , giving rise to an induced emf that must be overcome by a voltage  $V_{21} = \pm L_{21} \frac{di_2}{dt}$ , in order to keep  $i_1$  constant at its value  $I_1$ .

The work involved in

$$W_{21} = \int V_{21} I_1 dt = \pm L_{21} I_1 \int_0^{I_2} di_2 = \pm L_{21} I_1 I_2 \quad \dots \dots (ii)$$

In eq. (ii), the plus sign applies if  $I_1$  &  $I_2$  are such that their magnetic fields strengthen each other, and minus sign applies if their magnetic field



oppose each other.

At the same time a work  $W_{22}$  must be done in loop  $C_2$  in order to counteract the induced emf as  $i_2$  is increased from 0 to  $I_2$ .

$$W_{22} = \frac{1}{2} L_2 I_2^2 \quad \dots \dots \dots (iii)$$

The total amount of work done in raising the currents in loops  $C_1$  and  $C_2$  from zero to  $I_1$  and  $I_2$ , respectively, is then the sum of  $W_1$ ,  $W_{21}$ , and  $W_{22}$ ;

$$W_2 = \frac{1}{2} L_1 I_1^2 \pm L_{21} I_1 I_2 + \frac{1}{2} L_2 I_2^2$$

which is the energy stored in the magnetic field of the two coupled current carrying loops.

\*\* For a current  $I$  flowing in a single inductor with inductance  $L$ , the stored magnetic energy is;

$$W_m = \frac{1}{2} L I^2 \quad (J).$$



5-12.1 Determine the force between two current carrying conductors.

(24)

Ans:- Let us consider an element of conductor  $d\vec{r}$  with

$N$  = number of charge carriers per unit volume (number density)

$e$  = charge of each carrier

$S$  = cross sectional area of the wire

$\vec{u}$  = drift velocity

$I$  = current

the direction of  $\vec{u}$  and  $d\vec{r}$  is same.

Now, from Lorentz force equation we know

$$\vec{F}_m = q \vec{u} \times \vec{B}$$

In the current situation,

$$d\vec{F}_m = -NeS|d\vec{r}|\vec{u} \times \vec{B}$$

$$[\because q = NeSd\vec{r} \\ Sd\vec{r} = \text{volume}]$$

$$= -NeS|\vec{u}|d\vec{r} \times \vec{B}$$

since,  $-NeS|\vec{u}| = I$

$[I = \frac{dQ}{dt}]$  in time  $dt$   
charge move a distance  
 $= udt$   
So volume of charge crossing  
area  $S$ , volume  $= S udt$   
Number of charges  $= N S udt$



$$[\because I = \frac{dQ}{dt}]$$

In time  $dt$ , charges move a distance:

$$\text{distance} = u dt \quad [\because d = vt]$$

So, volume of charges crossing area  $S$ :

$$\text{Volume} = S u dt$$

$$\text{Number of charges} = N S u dt$$

$$\text{Total charge, } dQ = N (S u dt) e$$

$$I = \frac{dQ}{dt} = \frac{N e S u dt}{dt} = N e S u$$

$$\text{Then, } d\vec{F}_m = I d\vec{\ell} \times \vec{B} \quad (N)$$

The magnetic force on a complete (closed) circuit of contour  $C$  that carries a current  $I$  in a magnetic field  $\vec{B}$  is then

$$\vec{F}_m = I \oint_C d\vec{\ell} \times \vec{B} \quad (N).$$



When we have two circuits carrying currents  $I_1$  and  $I_2$ , respectively.

In the presence of magnetic flux  $\vec{B}_{12}$ , which was caused by the current  $I_1$  in  $C_1$ , the force  $\vec{F}_{12}$  on circuit  $C_2$  can be written as,

$$\vec{F}_{12} = I_2 \oint_{C_2} d\vec{l}_2 \times \vec{B}_{12} \quad \dots (i)$$

where,  $\vec{B}_{12}$  is, from the Biot-Savart law

$$\vec{B}_{12} = \frac{\mu_0 I_1}{4\pi} \oint_{C_1} \frac{d\vec{l}_1 \times \vec{r}_{12}}{r_{12}^2} \quad \dots (ii)$$

[ $\because F_{12}$  on  $C_2$  by  $C_1$  is  $\vec{B}_{12}$  on  $C_2$  by  $I_1$ ]

Combining Eqs. (i) & (ii), we obtain

$$\vec{F}_{12} = \frac{\mu_0}{4\pi} I_2 I_1 \oint_{C_2} \oint_{C_1} \frac{d\vec{l}_2 \times (d\vec{l}_1 \times \vec{r}_{12})}{r_{12}^2} \quad (iii)$$

which is Ampere's law of force between two current-carrying circuits.